Fowler Flaps Mechanism

Term Project

2103-322 Design of Mechanism – 2569

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Background and Significance of Mechanism

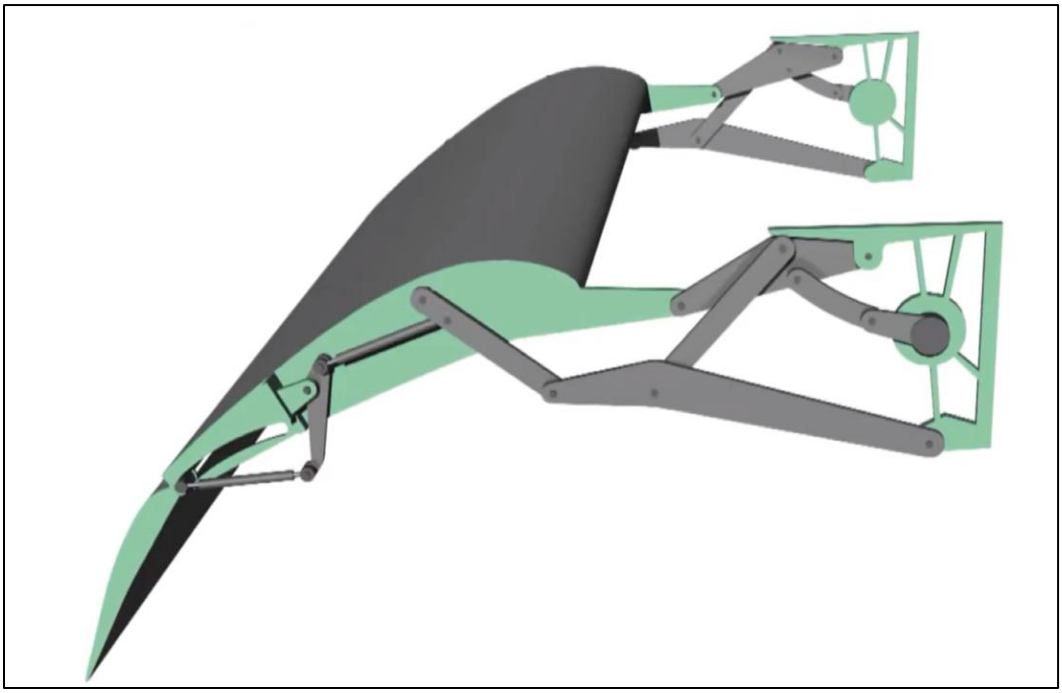
The Fowler Flap mechanism is a critical high-lift device used in aeronautical engineering to enhance lift generation during take-off and landing phases. Unlike conventional flaps, the Fowler Flap uniquely combines two distinct motions: rearward translation to expand the total wing surface area and downward deflection to increase wing camber. This dual-action motion significantly raises the overall maximum lift coefficient (C_L).

Because this mechanism acts as both a motion generator (extending along a track) and a function generator (deflecting the flap profile), it presents an ideal system for kinematic and dynamic modeling in the 2103-322 Design of Mechanism course. This project investigates and analyzes the Fowler Flap mechanism inspired by the operational concepts demonstrated in the reference video.

Objectives

1. Kinematic and DoF Analysis: Construct the kinematic diagram, classify links and joints, evaluate the degrees of freedom (DoF) using Kutzbach's criterion, and define the operational workspace.
2. 3D CAD Design and Assembly: Create a detailed 3D solid model and assembly of the Fowler Flap mechanism components (links, rollers, track profile, and flap body) using Autodesk Fusion 360 with exact geometric constraints.
3. Symbolic Mathematical Modeling: Formulate vector-loop equations in MATLAB to analyze the mechanism's position.
4. Multibody Simulation: Build 3D computer models using Fusion 360 and Simscape Multibody to generate a motion video (Mechanism.avi) and a high-resolution render (Render.jpg) showing the operating range.

Selected Mechanism Description



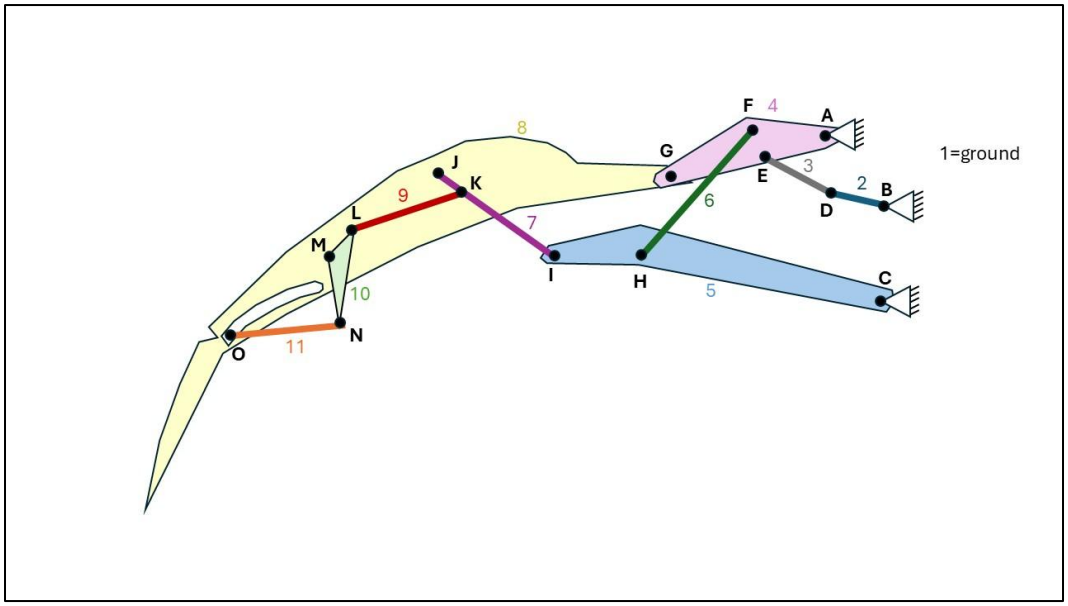
Reference: How Fowler Flaps Work

The mechanism selected for this project is the Fowler Flap Mechanism, a high-lift trailing-edge device engineered to alter wing geometry during critical flight phases.

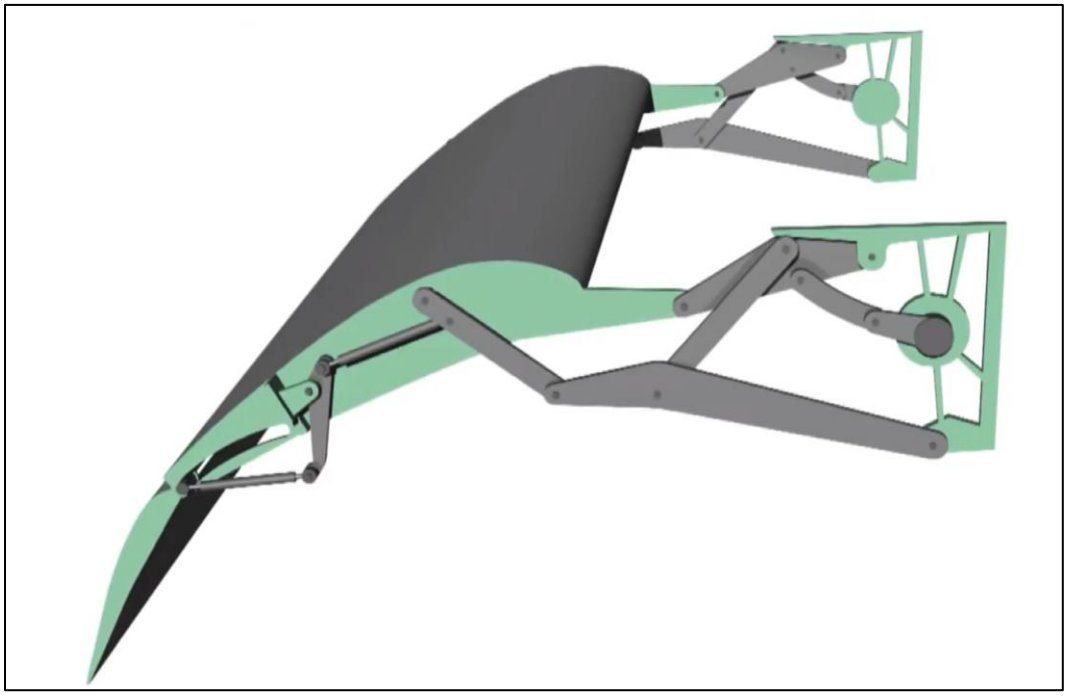
Mechanical Working Principle

- Motion Generator Function: The system acts as a motion generator, guiding the flap body along a predetermined trajectory that combines rearward translation with downward pitch.
- Input-to-Output Transmission: A single drive input (such as a rotary actuator or push-pull rod) powers the mechanical links and roller assembly, converting continuous input motion into a smooth, coupled dual-axis output displacement.
- Operational Configurations:
    - Stowed Position (Cruise): The flap is fully retracted inside the wing envelope, maintaining a clean and aerodynamic airfoil profile with low drag.
    - Partially Deployed (Take-off): The flap extends rearward to expand the total wing surface area, generating maximum lift with minimal added drag.
    - Fully Deployed (Landing): The flap reaches maximum extension and tilts downward at its maximum deflection angle, significantly increasing wing camber and aerodynamic drag for deceleration.

Kinematic Diagram



Kinematics Diagram



Reference

Joints Description

As in Kinematics Diagram, each joint represents by

Joint A: Pin joints connect ground with link 4

Joint B: Pin joints connect ground with link 2

Joint C: Pin joints connect ground with link 5

Joint D: Pin joints connect link 2 with link 3

Joint E: Pin joints connect link 3 with link 4

Joint F: Pin joints connect link 4 with link 6

Joint G: Pin joints connect link 4 with link 8

Joint H: Pin joints connect link 5 with link 6

Joint I: Pin joints connect link 5 with link 7

Joint J: Pin joints connect link 7 with link 8

Joint K: Pin joints connect link 7 with link 9

Joint L: Pin joints connect link 9 with link 10

Joint M: Pin joints connect link 8 with link 10

Joint N: Pin joints connect link 10 with link 11

Joint O: Curve slots connect link 11 to curve slot in link 8

DOF Analysis

From the diagram we see that Link = 11, Joint 1 = 14, Joint 2 = 1

From Kutzbach Criterion for Planar Mechanisms

$DOF = 3(L-1) - 2J1 - J2 = 3(11-1) - 2(14) - 1$

$\therefore DOF = 1$

Verification and Physical Significance

The resulting mobility of $M = 1$ confirms that the selected Fowler Flap configuration is a fully constrained (determinate) mechanism, which perfectly aligns with its operational requirements:

1. Single-Input Actuation: A mobility of 1 means that the entire complex motion profile of the flap—combining backward translation (surface area expansion) and downward tilt (camber change)—can be fully driven and precisely controlled by a single actuator (e.g., one rotary drive shaft or linear hydraulic cylinder).
2. Kinematic Determinacy: Every intermediate position, velocity, and acceleration of the flap body is uniquely determined by the input angle/displacement of the drive mechanism. This eliminates uncontrolled play, chatter, or secondary unconstrained movement during flight.

3D CAD Design and Assembly

1. CAD Design & Parameter Standardization

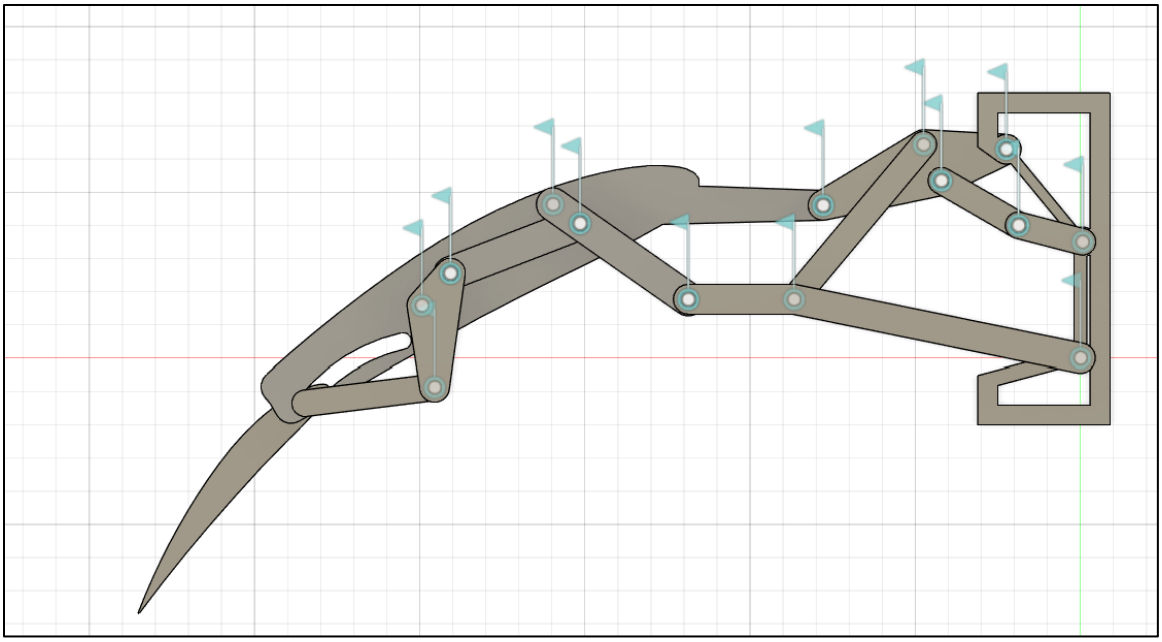
To ensure seamless integration with Simscape Multibody and eliminate numerical inconsistencies during kinematic and dynamic simulations, the 3D model of the Fowler Flap mechanism was created with standardized parametric geometry in Autodesk Fusion 360:

- Standardized Thickness: All structural links and moving components are extruded to a uniform thickness of 5 mm. This maintains a clean planar geometry while providing clear spatial separation between overlapping links in 3D space.
- Standardized Pin Joint Interfaces: All revolute joint connections are standardized with 5 mm circular pins (bosses) and matching 5 mm holes.
- A4 Workspace Envelope Constraint: The overall mechanism assembly is strictly dimensioned to operate within an A4-sized boundary envelope (210×297 mm). This ensures geometric consistency across the analytical calculations, Multibody simulation environment, and physical limits.

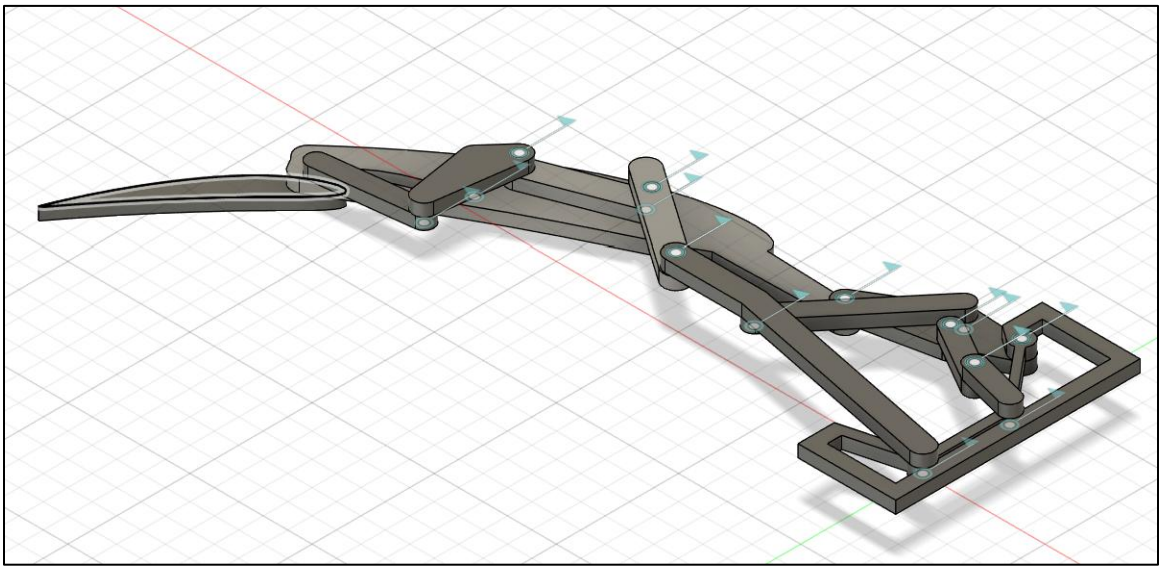
2. Engineering Rationale for Multibody Simulation

The simplified CAD approach was intentionally chosen specifically for compatibility with Simscape Multibody:

1. Precise Reference Frame Definition: Standardizing all joints as concentric 5 mm cylinders/holes provide precise centerlines for placing Rigid Transform coordinate frames (CS) at each joint axis.
2. Simplified Inertia and Mass Properties: Using uniform extrusions simplifies automatic calculation of mass centers (CG), moments of inertia, and mass matrices in Simscape, reducing potential dynamic simulation errors.
3. Prevention of Spatial Interference: The 5 mm standardized link thickness allows easy offset stacking along the Z-axis, preventing body collisions during multibody motion visualization while preserving planar 2D kinematic behavior.



Top view of CAD Design



Isometric view of CAD Design

For the Joint displacement and initial position

A = (0 mm, 0 mm)

B = (0.61 mm, 34.99 mm)

C = (-22.46 mm, 62.98 mm)

AF = 25.09 mm

AG = 58 mm

CH = 88.5 mm

FH = 61 mm

CI = 120.03 mm

GJ = 81.76 mm

IJ = 50 mm

KI = 40 mm

GM = 125.11 mm

ML = 13 mm

KL = 42 mm

MN = 24.75 mm

NO = 39.5 mm

Radius curve slot = 50 mm

Arcspan curve slot = 34.94°

Symbolic Mathematical Modeling

The kinematic modeling methodology and 2D animation implementation developed in MATLAB for a multi-bar planar linkage mechanism. The system comprises 11 moving links (Link 2 through Link 11) supported by three fixed ground pivots and coupled at the output stage through a pin-in-arc-slot constraint to guide the end-effector tip T along a prescribed curvilinear trajectory.

The linkage is anchored to three fixed ground pivots: O2(0.61, 34.99) mm, O4(-22.46, 62.98) mm, and the coordinate origin OC(0, 0) mm. Power transmission flows sequentially through four functional kinematic stages:

- Primary Drive Stage (Links 2–4): Input crank Link 2 rotates about ground pivot O2, driving coupler Link 3 and ternary rocker Link 4 pivoted at O4 (carrying rigid extension points F and G).
- Intermediate Transmission Stage (Links 5–8): Link 6 connects point F to ground-pivoted Link 5 (at OC), which drives Link 7 and ternary Link 8 (G-J-M) supporting the machined arc slot.
- Distal Articulation Stage (Links 9–10): Link 9 connects extension point K on Link 7 to ternary Link 10 (L-M-N), which pivots relative to Link 8 at joint M.

Arc Slot & Output Stage (Arc Slot & Link 11): Pin O of Link 11 is constrained to slide inside the 50 mm radius arc slot fixed on Link 8 while articulating at joint N on Link 10, guiding the output tip T.

Mathematical & Kinematic Methodology

The solver employs a sequential closed-form vector-loop position analysis. At every discrete crank angle $\theta 2$, joint coordinates are resolved analytically in upstream-to-downstream order without iterative numerical root-finding.

- Circle-Circle Intersection Solver (circInt): All binary revolute joints connecting two rigid links (joints E, H, J, L, and O) are computed as the geometric intersection of two circles centered at A (radius rA) and B (radius rB):
    - Analytical Formulation in circInt(A, rA, B, rB, s)

        Center-to-center distance & unit vector: $d = ||B - A||$, $u = (B - A) / d$

        Axial projection along line of centers: $a = (rA^2 - rB^2 + d^2) / (2d)$

        Perpendicular offset height: $h = \text{sqrt}(\max(rA^2 - a^2, 0))$

        Selected intersection point: $P = A + a \cdot u + s \cdot h \cdot [-u_y, u_x]$, where $s \in \{+1, -1\}$ prevents branch flipping

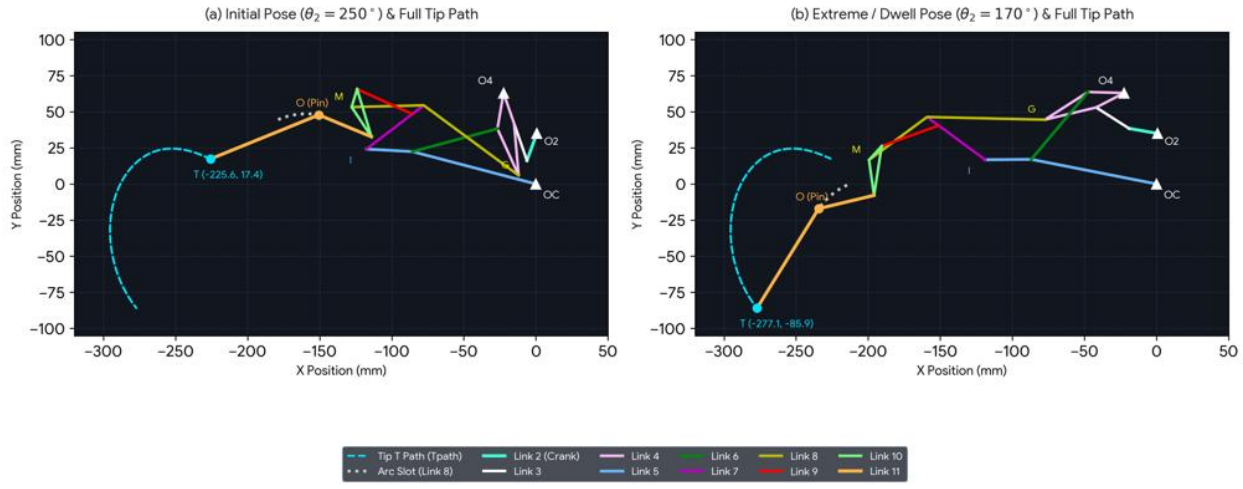
- Local Coordinate Frame Mapping for the Arc Slot on Link 8: Because the curved guide slot is machined directly into Link 8, its center of curvature (Cc) and angular start boundary translate and rotate rigidly with Link 8 throughout the cycle. This is handled in two steps:

  Step 1: Reference Pose Calibration (θ2 = 170°): Using the world coordinates measured in Fusion360 (cWorldRef), the solver transforms the slot center into Link 8's local coordinate frame relative to pivot G and orientation θ8 via the inverse rotation matrix R(-θ8), storing constants cLocal and arcStartLoc.

  Step 2: Frame-by-Frame World Update: At each simulation step, the slot center is mapped back into global coordinates. The slot profile is rendered using 40 angular points across the 34.94° span, and pin O is solved as the intersection between circle (N, NO) and slot circle (Cc, Rarc).

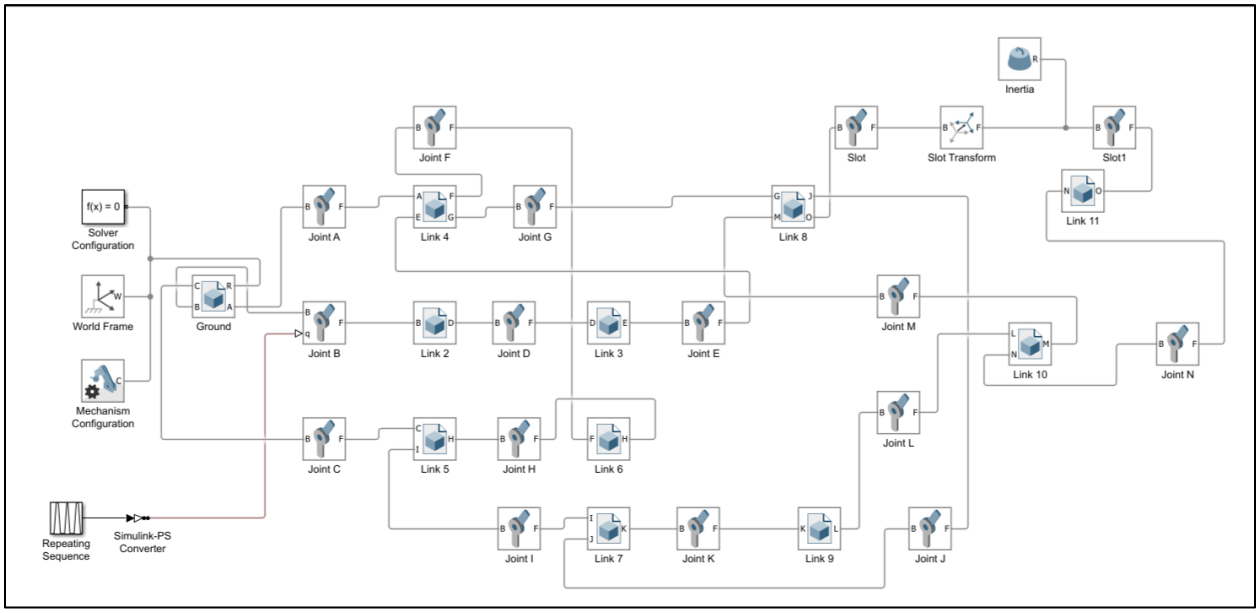
Simulation Results

Evaluating the closed-form kinematic model across all 352 animation frames confirms continuous, defect-free assembly of all 11 links throughout the forward, dwell, and return phases



The MATLAB kinematic simulation accurately models the 11-bar planar linkage and its body-fixed arc-slot constraint. Transforming the SolidWorks reference geometry into Link 8's local coordinate frame ensures exact geometric synchronization throughout motion, and the resulting 352-frame animation verifies that end-effector tip T achieves its prescribed curvilinear trajectory cleanly and without mechanical interference.

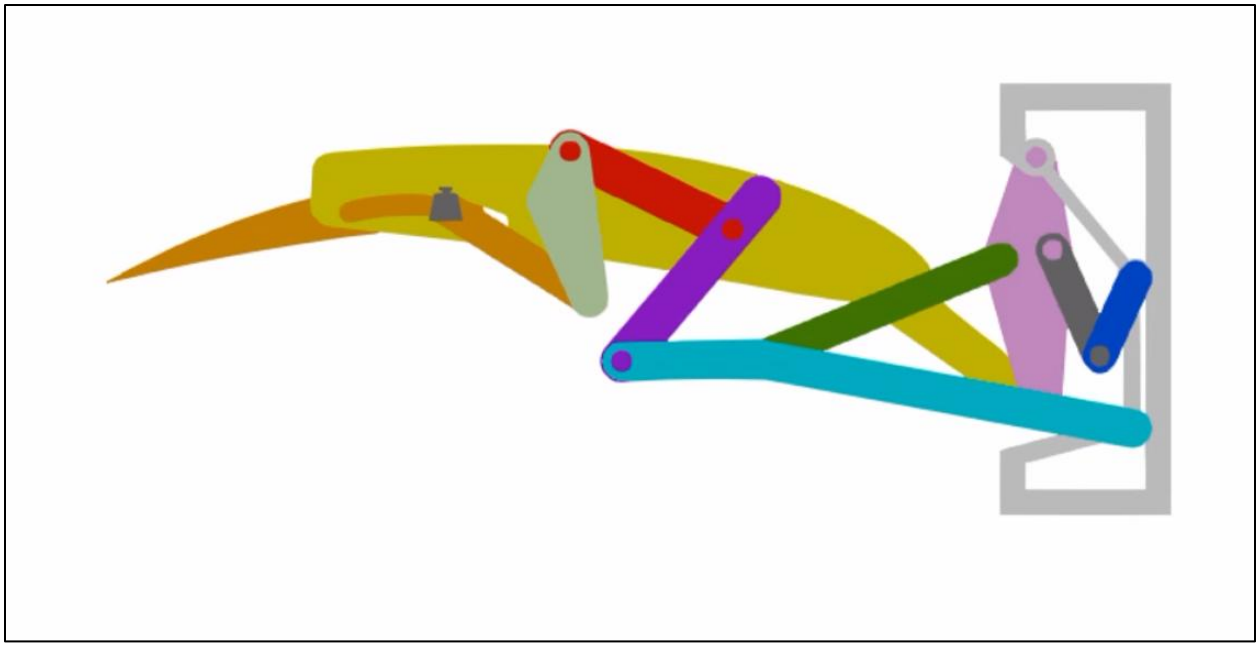
Multibody Simulation



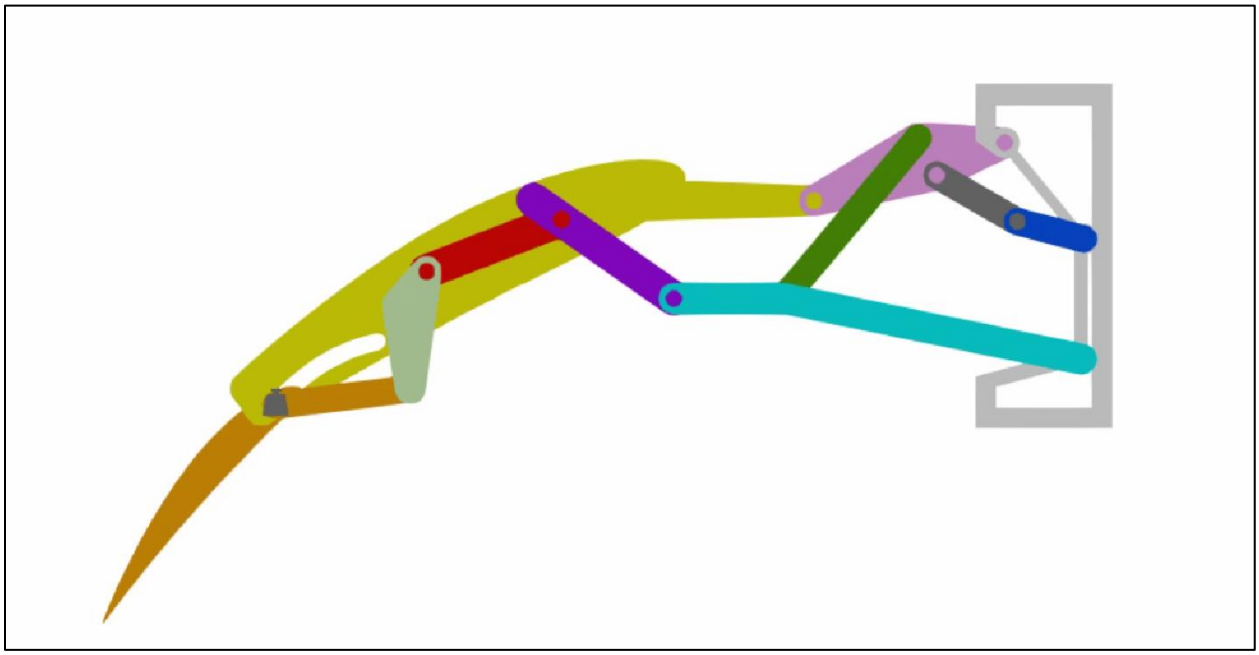
Multibody Simulink block diagram

The dynamic and kinematic simulation of the Fowler Flap mechanism is constructed in **Simscape Multibody** by connecting rigid body blocks, joint primitives, and physical drive signals:

- **Global Environment & Solver Setup:** The World Frame establishes the global 3D spatial reference system, while Mechanism Configuration defines gravity and global reference parameters. The Solver Configuration block applies stiff numerical ODE solvers to solve the closed-loop differential-algebraic equations (DAEs) governing the mechanism's constrained motion.
- **Input Actuation Mechanism:** System actuation is driven at **Joint B** using a prescribed position trajectory generated by a Repeating Sequence block. The signal is converted into a physical domain input via the Simulink-PS Converter to drive the joint motion continuously through the operational cycle.
- **Linkage & Joint Topology:** Structural elements (represented by Link 2 through Link 11) are connected in series-parallel loops via revolute joints (Joint A through Joint N). The rearward translational motion along the guide path is governed by dedicated track slider transforms (Slot, Slot Transform, and Slot1).
- **Motion Output:** As Joint B rotates, the interconnected linkage network forces the flap body to translate along the track while simultaneously rotating downward, reproducing the exact 1-DOF dual-action motion profile of a Fowler Flap.



3D Mechanics Explorer visualization showing the mechanism in its fully retracted (stowed) state, representing the baseline cruise configuration with minimal drag.



3D Mechanics Explorer visualization showing the mechanism at full stroke (fully deployed), demonstrating maximum rearward extension and downward angle deflection.